



- Q1-** How would you determine the standard electrode potential of the system $\text{Mg}^{2+} | \text{Mg}$?
- Q2-** Can you store copper sulphate solutions in a zinc pot?
- Q3-** Consult the table of standard electrode potentials and suggest three substances that can oxidise ferrous ions under suitable conditions.
- Q4-** Represent the cell in which the following reaction takes place
- $$\text{Mg(s)} + 2\text{Ag}^+(0.0001\text{M}) \rightarrow \text{Mg}^{2+}(0.130\text{M}) + 2\text{Ag(s)}$$
- Calculate its $E_{\text{(cell)}}^{\circ}$ if $E_{\text{(cell)}}^{\circ} = 3.17 \text{ V}$.
- Q5-** Calculate the equilibrium constant of the reaction:
- $$\text{Cu(s)} + 2\text{Ag(aq)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag(s)} \quad E_{\text{(cell)}}^{\circ} = 0.46\text{V}$$
- Q6-** The standard electrode potential for Daniell cell is 1.1V. Calculate the standard Gibbs energy for the reaction:
- $$\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$$
- Q7-** Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.
- Q8-** Calculate the emf of the cell in which the following reaction takes place:
- $$\text{Ni(s)} + 2\text{Ag}^+(0.002\text{M}) \rightarrow \text{Ni}^{2+}(0.16\text{M}) + 2\text{Ag(s)}$$
- Given that $E_{\text{(cell)}}^{\circ} = 1.05 \text{ V}$
- Q9-** The cell in which the following reaction occurs:
- $$2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{s})$$
- has $E_{\text{(cell)}}^{\circ} = 0.236\text{V}$ at 298 K.
- Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.
- Q10-** Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100Ω . If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520Ω , calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S/m .
- Q11-** The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of 5.55×10^3 diameter 1 cm and length 50 cm is ohm. Calculate its resistivity, conductivity and molar conductivity.
- Q12-** The molar conductivity of KCl solutions at different concentrations at 298 K are given below:
- | $c/\text{mol L}^{-1}$ | $\Lambda_m/\text{S cm}^2 \text{ mol}^{-1}$ |
|-----------------------|--|
| 0.000198 | 148.61 |
| 0.000309 | 148.29 |
| 0.000521 | 147.81 |
| 0.000989 | 147.09 |



Show that a plot between Λ_m and $c^{1/2}$ is a straight line. Determine the values of Λ_m^0 and A for KCl.

- Q13-** Calculate Λ_m^0 for CaCl_2 and MgSO_4 from the data given in above
- Q14-** Λ_m^0 for NaCl, HCl and NaAc are 126.4, 425.9 and 91.0 $\text{S cm}^2 \text{mol}^{-1}$ respectively, Calculate Λ^0 for HAC.
- Q15-** The conductivity of 0.001028 mol L^{-1} acetic acid is $4.95 \times 10^{-5} \text{ S cm}^{-1}$ Calculate its dissociation constant if Λ_m for acetic acid is $90.5 \text{ S cm}^2 \text{mol}^{-1}$.
- Q16-** Why does the conductivity of a solution decrease with dilution?
- Q17-** Suggest a way to determine the Λ_m value of water.
- Q18-** The molar conductivity of 0.025 mol L^{-1} methanoic acid is $46.1 \text{ S cm}^2 \text{mol}^{-1}$. Calculate its degree of dissociation and dissociation constant. Given $\lambda^0(\text{H}) = 349.6 \text{ S cm}^2 \text{mol}^{-1}$ and $\lambda^0(\text{HCOO}^-) = 54.6 \text{ S cm}^2 \text{mol}^{-1}$.
- Q19-** A solution of CuSO_4 is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode?
- Q20-** If a current of 0.5 ampere flows through a metallic wire for 2 hours, then how many electrons would flow through the wire?
- Q21-** Suggest a list of metals that are extracted electrolytically.
- Q22-** Consider the reaction: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ What is the quantity of electricity in coulombs needed to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$?
- Q23-** Write the chemistry of recharging the lead storage battery, highlighting all the materials that are involved during recharging.
- Q24-** Suggest two materials other than hydrogen that can be used as fuels in fuel cells.
- Q25-** Explain how rusting of iron is envisaged as setting up of an electrochemical cell.
- Q26-** Arrange the following metals in the order in which they displace each other from the solution of their salts. Al, Cu, Fe, Mg and Zn.
- Q27-** Given the standard electrode potentials.
 $\text{K}^+/\text{K} = -2.93\text{V}$, $\text{Ag}^+/\text{Ag} = 0.8\text{V}$
 $\text{Hg}^{2+}/\text{Hg} = 0.79\text{V}$, $\text{Mg}^{2+}/\text{Mg} = -2.37\text{V}$, $\text{Cr}^{3+}/\text{Cr} = -0.74\text{V}$
Arrange these metals in their increasing order of reducing power.



Q28- Depict the galvanic cell in which the reaction $\text{Zn(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Ag(s)}$ takes place. Further show:

- Which of the electrode is negatively charged?
- The carriers of the current in the cell.
- Individual reaction at each electrode.

Q29- Calculate the standard cell potentials of galvanic cell in which the following reactions take place:

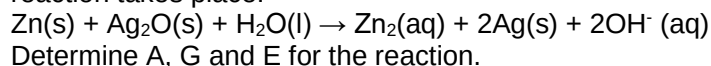
- $2\text{Cr(s)} + 3\text{Cd}^{2+}(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{Cd}$
- $\text{Fe}^{2+}(\text{aq}) + \text{Ag}(\text{aq}) \rightarrow \text{Fe}(\text{aq}) + \text{Ag(s)}$

Calculate the AG and equilibrium constant of the reactions.

Q30- Write the Nernst equation and emf of the following cells at 298 K:

- $\text{Mg(s)} \mid \text{Mg}^{2+}(0.001\text{M}) \parallel \text{Cu}^{2+}(0.0001\text{M}) \mid \text{Cu(s)}$
- $\text{Fe(s)} \mid \text{Fe}^{2+}(0.001\text{M}) \parallel \text{H}^+(1\text{M}) \mid \text{H}_2(\text{g})(1\text{bar}) \mid \text{Pt(s)}$
- $\text{Sn(s)} \mid \text{Sn}^{2+}(0.050\text{M}) \mid \text{H}^+(0.020\text{M}) \mid \text{H}_2(\text{g})(1\text{bar}) \mid \text{Pt(s)}$
- $\text{Pt(s)} \mid \text{Br}^-(0.010\text{M}) \mid \text{Br}_2(1) \parallel \text{H}^+(0.030\text{M}) \mid \text{H}_2(\text{g})(1\text{bar}) \mid \text{Pt(s)}$

Q31- In the button cells widely used in watches and other devices the following reaction takes place:



Q32- Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.

Q33- The conductivity of 0.20 M solution of KCl at 298 K is 0.0248 S cm^{-1} . Calculate its molar conductivity.

Q34- The resistance of a conductivity cell containing 0.001M KCl solution at 298 K is 1500 Ω . What is the cell constant if conductivity of 0.001M KCl solution at 298 K is $0.146 \times 10^{-3}\text{ S cm}^{-1}$,

Q35- The conductivity of sodium chloride at 298 K has been determined at different concentrations and the results are given below:

Concentration/M	0.001	0.010	0.020	0.050	0.100
$10 \times \kappa / \text{S m}^{-1}$	1.237	11.85	23.15	55.53	106.74

Calculate Λ_m for all concentrations and draw a plot between Λ_m and Find the value of Λ_m^0

Q36- Conductivity of 0.00241 M acetic acid is $7.896 \times 10^{-5}\text{ Sc m}^{-1}$ Calculate its molar conductivity. If Λ_m^0 for acetic acid is $390.5\text{ S cm}^2\text{ mol}^{-1}$, what is its dissociation constant?

Q37- How much charge is required for the following reductions:

- 1 mol of Al to Al?
- 1 mol of Cu^{2+} to Cu?
- 1 mol of MnO to Mn^{2+} ?



- Q38-** How much electricity in terms of Faraday is required to produce
(i) 20.0 g of Ca from molten CaCl_2 ?
(ii) 40.0 g of Al from molten Al_2O_3 ?
- Q39-** How much electricity is required in coulomb for the oxidation of
(i) 1 mol of H_2O to O_2 ?
(ii) 1 mol of FeO to Fe_2O_3 ?
- Q40-** A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using platinum a current of 5 amperes for 20 minutes. What mass of Ni is deposited at the cathode?
- Q41-** Three electrolytic cells A,B,C containing solutions of ZnSO_4 , AgNO_3 , and CuSO_4 respectively are connected in series. A steady current of 1.5 amperes was passed through them until 1.45 g of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited?
- Q42-** Using the standard electrode potentials given in Table 3.1. predict if the reaction between the following is feasible:
(i) $\text{Fe}^{3+}(\text{aq})$ and $\text{I}^-(\text{aq})$
(ii) $\text{Ag}^+(\text{aq})$ and $\text{Cu}(\text{s})$
(iii) $\text{Fe}^{3+}(\text{aq})$ and $\text{Br}^-(\text{aq})$
(iv) $\text{Ag}(\text{s})$ and $\text{Fe}^{3+}(\text{aq})$
(v) $\text{Br}_2(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$
- Q43-** Predict the products of electrolysis in each of the following:
(i) An aqueous solution of AgNO_3 , with silver electrodes.
(ii) An aqueous solution of AgNO_3 , with platinum electrodes.
(iii) A dilute solution of H_2SO_4 with platinum electrodes.
(iv) An aqueous solution of CuCl_2 with platinum electrodes.